INCH - POUNDS

MIL-DTL-28731/29C
w/AMENDMENT 1
17 January 2020
SUPERSEDING
MIL-DTL-28731/29C
25 May 2012

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, INSERT (INSULATOR), MALE RECTANGULAR, PLUG, POLARIZED, CENTER SCREW LOCK, FOR 75 REMOVABLE TYPE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28731.

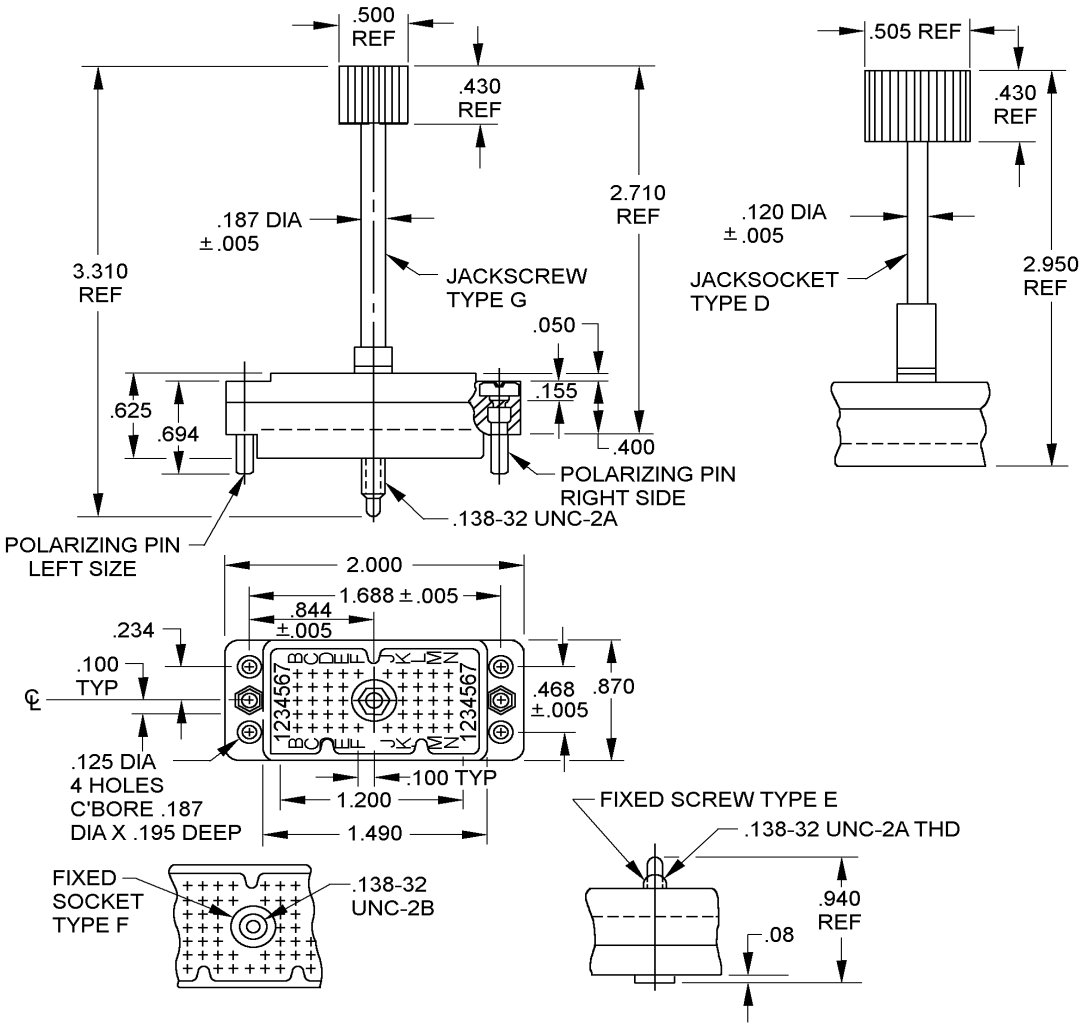


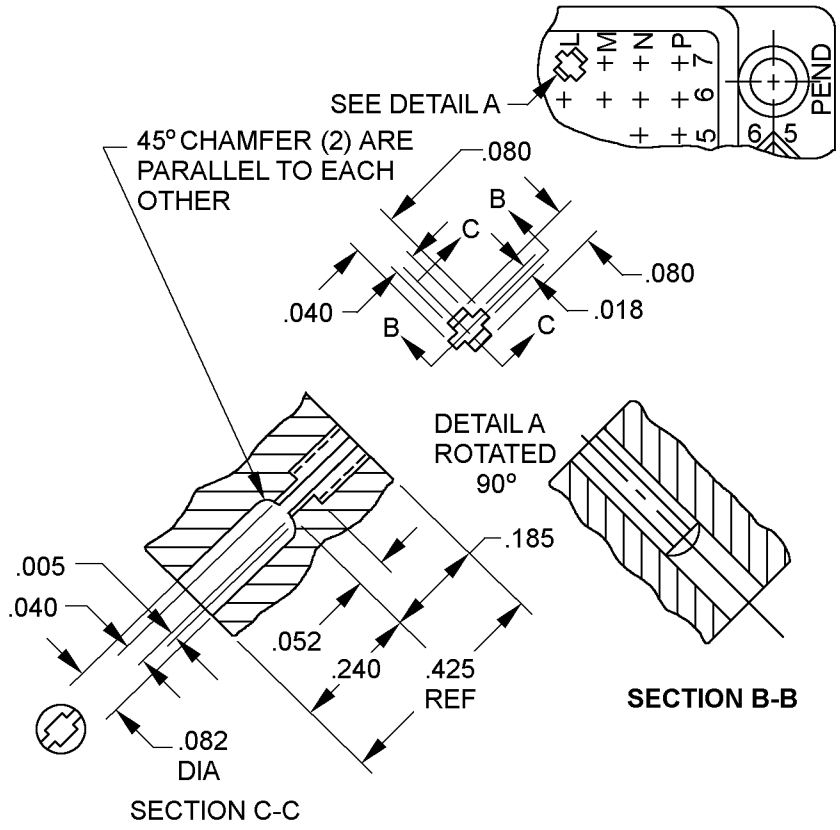
FIGURE 1. Connector, male rectangular, plug.

AMSC N/A

FSC 5935

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Inches	mm	Inches	mm
.005	0.13	.625	15.88
.050	1.27	.694	17.63
.08	2.0	.844	21.44
.100	2.54	.870	22.10
.120	3.05	.940	23.88
.125	3.18	1.200	30.48
.155	3.94	1.490	37.85
.187	4.75	1.688	42.88
.234	5.94	2.000	50.80
.400	10.16	2.710	68.83
.430	10.92	2.950	74.93
.468	11.89	3.310	84.07

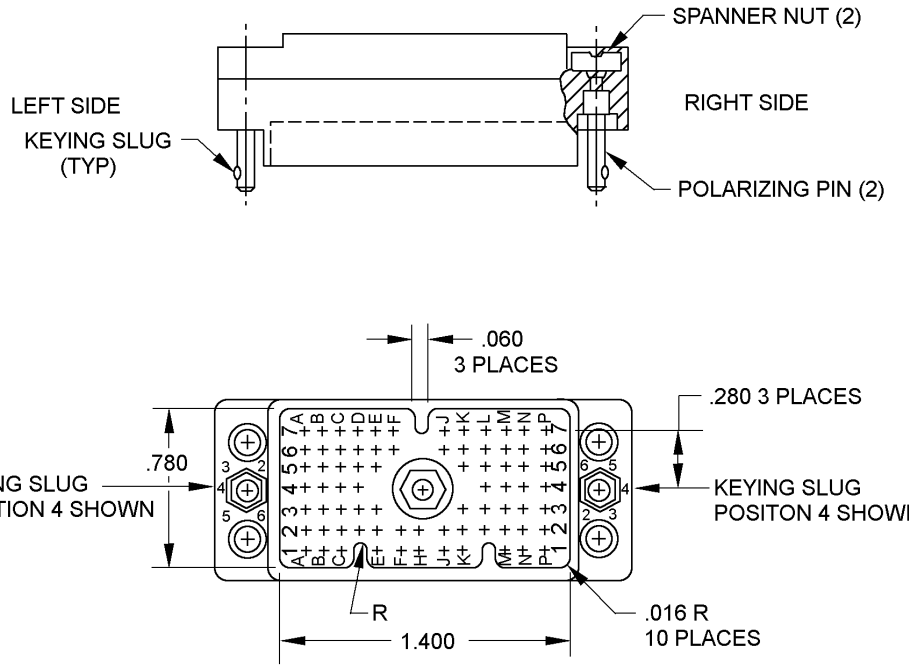


Inches	mm
.005	0.13
.018	0.46
.040	1.02
.052	1.32
.080	2.03
.082	2.08
.185	4.70
.240	6.10
.425	10.80

Configuration A – Contact hole detail

FIGURE 1. Connector, male rectangular, plug – Continued.

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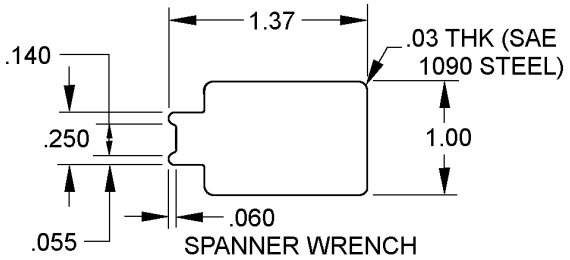
POLARIZING KEY MATING CODE	
P (MALE) Position	R (FEMALE) Position
4	1
5	2
6	3
1	4
2	5
3	6

Inches	mm
.016	0.41
.03	0.8
.055	1.4
.060	1.52
.140	3.56
.250	6.35
.280	7.11
.780	19.81
1.00	25.4
1.37	34.8
1.400	35.56

POLARIZING POSITION CHANGE

Keying position on both polarizing pins can be rotated to desired position by loosening spanner nut (counter-clockwise), resetting pin in one of six positions then retightening spanner nut (clockwise).

NOTE: Overtightening of spanner nut can fracture connector. Spanner keying wrench shown below.



CONFIGURATION OPTIONAL
DIMENSIONS REF. ONLY

Configuration B – Connector polarization guide (layout).

FIGURE 1. Connector, male rectangular, plug – Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Tolerances between any two contact centers shall be ± .004 (0.10 mm).
4. Flatness, squareness, and parallelism shall be within dimensional tolerances.
5. Contact hole details shall be in accordance with configuration B.
6. Mating connector information:
   M28731/29-0001 mate with M28731/30-0003 and -0005.
   M28731/29-0002 mate with M28731/30-0004 and -0005.
   M28731/29-0003 and -0005 mate with M28731/30-0001.
   M28731/29-0004 and -0006 mate with M28731/30-0002.
7. Polarization information: Polarization pins are provided for indexing to preclude intermixing of adjacent connectors. By using a standard type spanner wrench, polarization pins can be rotated to desired position. For parts supplied directly to the government, polarization will be in the 1-1 position. For other polarization positions see connector polarization guide configuration B (see marking, note 13d). For direct government acquisition a tool is supplied in the unit package for polarization socket adjustment.
8. Contact and terminal designations: Contacts and terminals shall be designated on the insulator using letters in the location shown on figure 1.
9. Tolerance shall be ± .010 (0.25 mm) unless otherwise specified.
10. Patent notice: Patent No. 3,675,185 owned by Elco Corporation. The government has a royalty-free license under the above patent for the benefit of manufacturers of items called for in this specification, either for the government or for use in equipment to be delivered to the government.
11. For direct government acquisition crimp contacts M28731/35-0001 (see table I) are furnished in a plastic bag in each unit package (see marking, note 13a).
12. For direct government acquisition wrappost contacts are M28731/36-0001 (see table I) and are factory installed in connector body (see marking, note 13a).
13. Marking:
    a. Connectors supplied directly to the government shall be marked with the military Part or Identifying Number (PIN) (see table I). Example: M28731/29-0001.
    b. Connectors supplied with contacts to government end users shall be marked with the military PIN (see table I). Example: M28731/29-0001.
    c. Connectors supplied without contacts to government end users shall be marked as follows: Example: M28731/29E (screw or socket type from table I). The user shall complete the remaining portion of the military PIN as shown in table I based upon contact type he assembles. Example: M28731/29E-0004.
    d. Polarization position marking is not required unless otherwise specified. When required polarization marking shall be in accordance with the connector polarization guide (see configuration B). The marking shall appear following the PIN.
14. Applicable contact insertion and removal tools:
    Insertion - MIL-I-81969/5-01
    Removal - MIL-I-81969/4-02
15. Rectangular shield: Shall conform to MIL-DTL-28731/19, PIN M28731/19-0003 through M28731/19-0006 and -0015, -0016 for type A and B hardware.
16. Recommended hardware (not furnished) for panel mounting: (4) .112-40-UNC-2A (length to suit) bolt with No. 4 washer and mating nut.

FIGURE 1. Connector, male rectangular, plug – Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Mating and unmating forces (torque): 96 inch-ounces, maximum.

Dielectric withstanding voltage: 1,000 volts rms (sea level).

Mechanical shock (specified pulse): Applicable.

Contact retention: 6 pounds.

Contact separation force: 1 ounce, minimum.

Contact current rating: 5 amperes, 22 AWG wire.

PIN: See table I.

TABLE I. PIN information.

Military PIN	Actuating hardware	Polarizing pin position		Contact type
		Left side	Right side	
M28731/29-0001	Jackscrew type G	4	4	Crimp, MIL-DTL-28731/35
M28731/29-0002	Jack socket type D	4	4	Crimp, MIL-DTL-28731/35
M28731/29-0003	Fixed socket type F	4	4	Crimp, MIL-DTL-28731/35
M28731/29-0004	Fixed screw type E	4	4	Crimp, MIL-DTL-28731/35
M28731/29-0005	Fixed socket type F	4	4	Wrappost, MIL-DTL-28731/36
M28731/29-0006	Fixed screw type E	4	4	Wrappost, MIL-DTL-28731/36

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-28731, this document references the following:

MIL-DTL-28731/19
MIL DTL-28731/35
MIL-DTL-28731/36

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2020-016)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – AS, MC, OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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